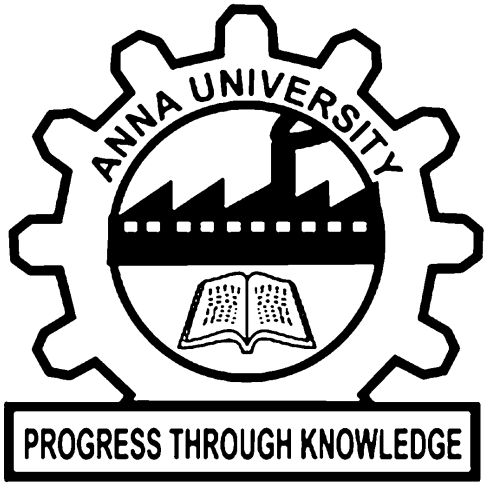
CLOUD COMPUTING

A SUMMER INTERNSHIP REPORT

Submitted by

BARATH R - 810023205022



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING, BIT CAMPUS, ANNA UNIVERSITY, TIRUCHIRAPPALLI – 620 024

UNIVERSITY COLLEGE OF ENGINEERING, BIT CAMPUS, ANNA UNIVERSITY, TIRUCHIRAPPALLI – 620 024

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that Summer internship project report **CLOUD COMPUTING** is the bonafide work of **BARATH R (810023205022)** has satisfactory completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.Tech INFORMATION TECHNOLOGY** during the academic year 2026-2027.

FACULTY INCHARGE **HEAD OF THE DEPARTMENT**

ABSTRACT

This report describes a 30-day internship in Cloud Computing undertaken at Fantasy Solution, Trichy, an IT solution and service provider. The internship introduced the service models (IaaS, PaaS and SaaS), the deployment models (public, private and hybrid) and the virtualization concepts on which modern cloud platforms are built.

Over the 30 days the work moved from fundamentals to hands-on practice with leading providers such as AWS, Microsoft Azure and Google Cloud. It covered virtual machines and containers, cloud storage, networking with Virtual Private Clouds, identity and access management, auto-scaling and load balancing, database and platform services, security and compliance, DevOps pipelines, Kubernetes, Infrastructure as Code, monitoring, disaster recovery, serverless computing, cost optimization and data warehousing.

Core Outcome: The internship ended with a mini project and a capstone project in which a scalable, secure and cost-aware cloud architecture was designed and documented, combining the tools and practices learned during the training.

The report presents the company profile, the work carried out, supporting diagrams and tables, the learnings gained and the conclusion, in a structured chapter format.

TABLE OF CONTENTS

Chapter	Title / Section Description	Page No.
--	Title Cover Page & Bonafide Certificate	1 – 2
--	Abstract & Table of Contents	3 – 4
1	Introduction to Internships	5
2	Purpose of an Internship	5 – 6
3	Company Profile: Fantasy Solution, Mission and Vision	6 – 8
4	Internship Overview, Work Plan and Tools	8 – 10
5	Cloud Computing Fundamentals and Architecture	10 – 12
6	Compute, Storage and Networking Services	12 – 14
7	Identity, Security, Encryption and Compliance	14 – 16
8	Scalability, Availability and Disaster Recovery	16 – 18
9	Platform Services, Databases and Hybrid Cloud	18 – 19
10	Cloud Cost Management and Optimization	19 – 20
11	DevOps, CI/CD and Infrastructure as Code	21 – 22
12	Microservices, Kubernetes and Serverless Computing	22 – 24
13	Cloud Migration, Monitoring and Logging	24 – 25
14	APIs, Data Warehousing and Application Frameworks	25 – 27
15	Mini Project and Capstone Project	27 – 28
16	Learnings and Outcomes	28 – 29
17	Conclusion and References	29 – 30

1. INTRODUCTION

An internship is a professional learning experience that offers meaningful, practical work related to a student's field of study or career interest. An internship gives a student the opportunity for career exploration and development, and to learn new skills. It offers the employer the opportunity to bring new ideas and energy into the workplace, develop talent and potentially build a pipeline for future full-time employees. It is an official program offered by organisations to help train and provide work experience to students and recent graduates.

Although the idea of working as an intern has been around for a while, it has undergone significant change. The early internships were performed by labourers who took on young people and trained them in their craft or profession. The trainee would consent to work for the labourer for a set period of time in return for being taught a skill. Even then, the goal of an internship—or better still, an apprenticeship—was to acquire new skills in order to be able to find employment in the future.

About this report: This report documents a 30-day Cloud Computing internship completed at Fantasy Solution, Trichy. It describes the company, the technologies studied and the practical work done, and closes with the learnings and conclusion drawn from the experience.

2. PURPOSE OF AN INTERNSHIP

Internships also often allow you to gain feedback from employers or mentors who already have experience in your desired industry. An internship can provide unique opportunities for learning outside of academic settings. It can expose you to new tasks and help you learn goal-specific skills to complete those tasks. Internships also give you experience with technology, people and projects that may relate to your career goals. Having a willingness to learn may help you optimize your internship experience.

There are a few reasons why an individual might decide to apply for an internship. Job listings often state that they prefer candidates with educational and job experience. If you're new to the workforce or attending school, you may consider looking for an internship to gain the experience required for most entry-level positions. An internship can give you an authentic experience in a job role by providing you with an

introductory experience to a career path, its duties and daily operations. If you enjoy your internship, this might indicate that your career is on the right path.

Internships can give you workplace experience before you actually enter the workforce. They also may assist you with developing additional skills to list on your resume, which can emphasize your value as a candidate.

3. COMPANY PROFILE

3.1 Company Address and Contact

Detail	Information
Company	Fantasy Solution
Address	No 16, Samnath Plazza, Third Floor, Near BG Naidu Sweets, Melapudur, Trichy-1.
Contact	9043095535
Landline	0431-4971630
Website	www.fantasysolution.in

3.2 Company Profile

Fantasy solution as a leading IT solution and service provider, provides innovative information technology - enabled solutions and services to meet the demands arising from social transformation, shaping new life styles for individuals and creating values for the society.

Focusing on software technology, Fantasy solution provides industry solutions and product engineering solutions, related software products & platforms, and services, through seamless integration of software and services, software and manufacturing, as well as technology and industrial management capacity.

Fantasy solution helps industry customers establish best practices in business development and management. The fantasy solution serves include real time projects, web designing, web hosting, software development and training etc, in many of which, has a leading market share. Notably, Fantasy Solution has participated in the formulation of many national IT standards and specifications.

Fantasy solution has the world's leading product engineering capabilities, ranging from consultation, design, R&D, and integration to testing of embedded software, in the fields of automotive electronics, smart devices, digital home products, and IT products. The software provided by fantasy solution runs in a number of globally renowned brands.

Particularly offering the services that include application development & maintenance, ERP implementation & consulting, testing, performance engineering, software localization & globalization, IT infrastructures, BPO, IT education & training, etc. Sticking to its business philosophy and brand commitment of "Beyond Technology", fantasy solution is dedicated to providing innovative information technologies to drive the sustainable development of society, as well as becoming a company that is well recognized and respected by employees, shareholders, customers, and society.

3.3 Our Services

In this ever-changing environment, keeping a competitive edge means being able to anticipate and respond quickly to changing business conditions. Fantasy solution is a global software development company providing IT solutions to enterprises worldwide. Combining proven expertise in technology, and an understanding of emerging business trends, Fantasy delivers a range of software development solutions that includes e-business solutions, computer telephony, enterprise applications, professional web site design and development, product engineering, Electronic Health Records, CMS Software's, Payment Gateway solutions, Time and attendance tracking software's, Debt collection software's, Appointment Reminder Solutions, Medical Transcription Services etc. We study, design, develop, enhance, customize, implement, maintain and support various aspects of information technology.

We are a professionally recognized software development company having huge experience in developing custom software development and application development best match to your need and requirements. We have expertise in working with a variety

of customers from companies to individuals. Our successful assignments with client companies have established our reputation as superior providers of IT solutions. Services offered by Fantasy includes Software development in .NET, PHP and Java, Web designing & development, Mobile application development in Android platforms, MATLAB for image processing, Network Simulator (NS2), Data mining tools, Big data development using R tool and Python. We aim to carve a position in the forefront, and it is our continuing goal to gain the trust of our clients. Our Motto is to serve the purpose of our clients with perfection.

3.4 Mission

Fantasy Solutions' mission includes:

1. Providing high quality software development services, professional consulting and development outsourcing that would improve our customers' operations;
2. Making access to information easier and securer (Enterprise Business);
3. Improving communication and data exchange (Business to Business);
4. Providing our customers with a Value for Money and
5. Providing our employees with meaningful work and advancement opportunities.

3.5 Vision

Vision: Fantasy Solutions is a leading IT company for Consulting Services and Deployment of best of breed Business Solutions to top tier domestic and international customers.

4. INTERNSHIP OVERVIEW AND WORK PLAN

4.1 Internship Overview

The internship was a 30-day Cloud Computing training programme. It began with the basic vocabulary of the cloud—service models, deployment models and virtualization—and gradually moved to practical work on compute, storage, networking and security, followed by DevOps, container orchestration, serverless computing and cost optimization. The programme closed with a mini project and a

capstone project in which the learned concepts were combined into complete cloud solutions.

4.2 Work Plan by Phase

Phase	Days	Focus Areas	Outcome
1. Foundations	1 – 5	Service and deployment models, architecture, VMs, containers, storage, networking	Working knowledge of core cloud services
2. Security & Scale	6 – 10	IAM, auto-scaling, load balancing, PaaS, DBaaS, provider comparison	Secure, scalable sample deployments
3. Governance & Operations	11 – 16	Cost management, hybrid cloud, security, compliance, CI/CD, encryption	Policies, pipelines and key management
4. Cloud-Native & Automation	17 – 24	Microservices, migration, Kubernetes, IaC, monitoring, disaster recovery, serverless	Automated, observable, resilient systems
5. Integration & Projects	25 – 30	Mini project, cost optimization, APIs, data warehousing, frameworks, capstone	Documented architectures and capstone solution

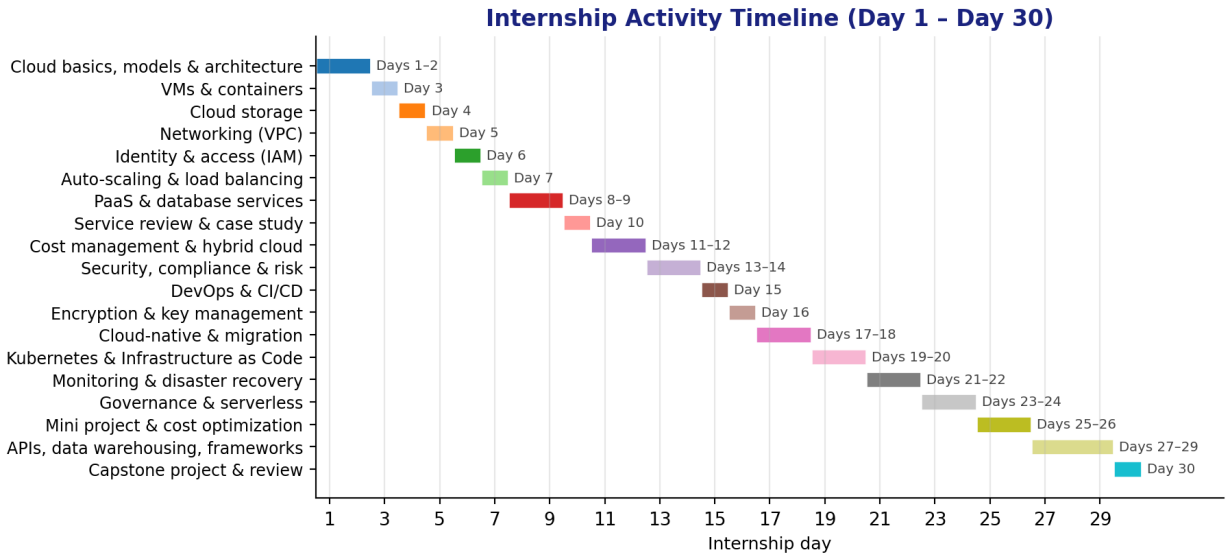


Figure 4.1: Internship Activity Timeline (Day 1 to Day 30)

4.3 Platforms and Tools Used

Category	Platforms / Tools
Cloud providers	Amazon Web Services (AWS), Microsoft Azure, Google Cloud
Compute & containers	Virtual machines, Docker, Kubernetes, AWS Lambda, Azure Functions
Storage & databases	Amazon S3, Google Cloud Storage, Amazon RDS, Firebase, Redshift, BigQuery
Security	IAM, multi-factor authentication, security groups, AWS KMS, Azure Key Vault
DevOps & IaC	GitHub Actions, AWS CodePipeline, Terraform, AWS CloudFormation
Monitoring & cost	Prometheus, Grafana, AWS CloudWatch, Cost Explorer, Azure Cost Management
PaaS & frameworks	Heroku, Google App Engine, AWS Amplify, Azure App Service

5. CLOUD COMPUTING FUNDAMENTALS

5.1 Service Models: IaaS, PaaS and SaaS

The internship started with the basics of cloud computing and its three service models. In Infrastructure as a Service (IaaS) the provider supplies virtual servers, storage and networks and the customer manages the operating system and everything above it. In Platform as a Service (PaaS) the provider also manages the runtime and middleware, so developers only deploy code. In Software as a Service (SaaS) the complete application is delivered over the internet and the customer simply uses it.

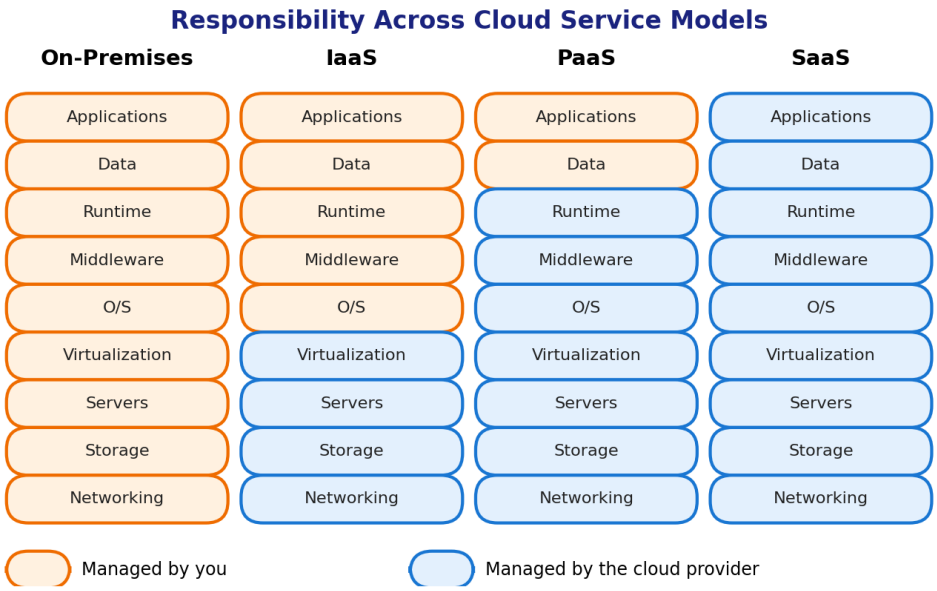


Figure 5.1: Responsibility Across Cloud Service Models

5.2 Deployment Models

Model	Description	Typical Use
Public cloud	Resources owned by a provider and shared by many customers over the internet.	Web applications, start-ups, variable workloads
Private cloud	Infrastructure dedicated to a single organisation, on-site or hosted.	Sensitive data, strict regulation
Hybrid cloud	A combination of private and public cloud connected to move data and applications.	Gradual migration, burst capacity
Multi-cloud	Services from more than one public cloud provider used together.	Avoiding vendor lock-in, best-of-breed services

5.3 Virtualization and Cloud Architecture

Virtualization is the technology that makes the cloud possible. A hypervisor divides a physical server into several isolated virtual machines, so that many customers can share the same hardware safely and resources can be created or removed within minutes. On Day 2 the study moved on to the architecture of a cloud platform: physical data centers grouped into regions and availability zones, a virtualization layer, and compute, storage and database services reached through load balancers and APIs, all protected by identity management and observed by monitoring and billing services.

5.4 Leading Cloud Service Providers

Capability	AWS	Microsoft Azure	Google Cloud
Virtual machines	EC2	Virtual Machines	Compute Engine
Object storage	S3	Blob Storage	Cloud Storage
Relational database	RDS	Azure SQL Database	Cloud SQL
Serverless functions	Lambda	Azure Functions	Cloud Functions
Containers (Kubernetes)	EKS	AKS	GKE
Data warehouse	Redshift	Synapse Analytics	BigQuery

Learning the equivalent services across the three providers made it easier to understand that the concepts are common even though the product names differ.

6. COMPUTE, STORAGE AND NETWORKING

6.1 Virtual Machines and Containers

The third day of the internship was spent on Virtual Machines (VMs) and containers. VMs were created and managed on a cloud platform, and their configuration and scalability options—instance size, storage volumes, security settings and images—were studied. Docker containers were then introduced and their role in microservices was understood.

Aspect	Virtual Machine	Container
Isolation	Full guest operating system per VM	Shares the host kernel; process-level isolation
Start-up time	Minutes	Seconds or less
Size	Gigabytes	Megabytes
Portability	Image tied to hypervisor format	Runs the same on any container host
Best for	Legacy apps, full OS control	Microservices, rapid scaling

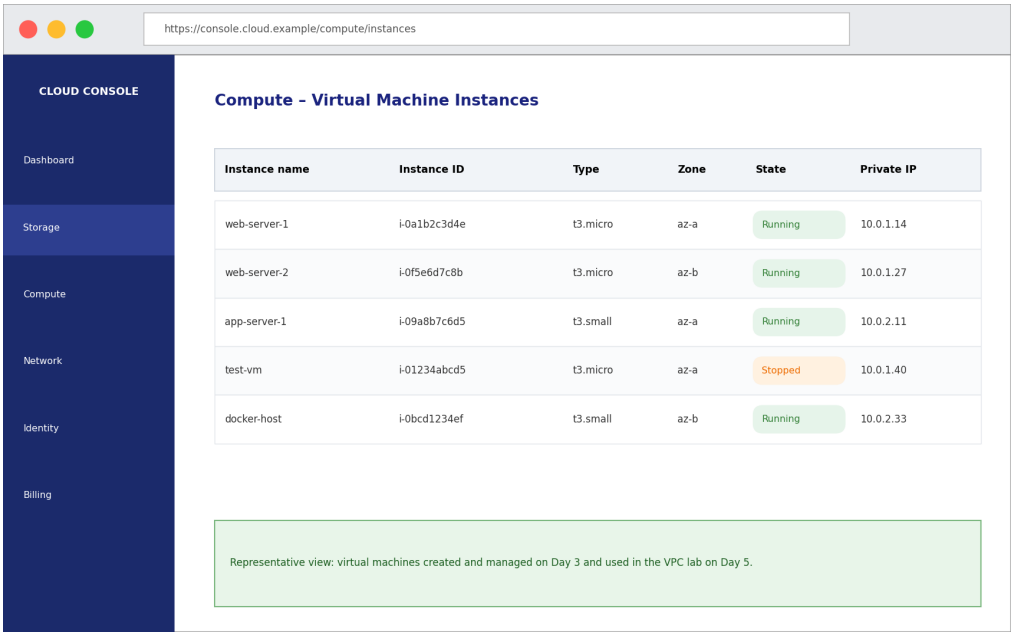


Figure 6.1: Virtual Machine Instances (representative console view)

6.2 Cloud Storage

Cloud storage services were studied in depth on Day 4 by comparing block, object and file storage. Amazon S3 and Google Cloud Storage were used to upload, manage and secure data, and data lifecycle policies and access control mechanisms were configured so that old data moves automatically to cheaper tiers or is deleted.

Type	How data is stored	Examples	Typical Use
Block	Fixed-size blocks attached to a VM like a disk	EBS, Persistent Disk	Operating systems, databases
Object	Objects with metadata in flat buckets, accessed by API	Amazon S3, Cloud Storage	Backups, media, static websites
File	Shared hierarchical file system over a network	EFS, Azure Files	Shared content, home directories

6.3 Cloud Networking and Virtual Private Cloud

Day 5 was dedicated to cloud networking concepts such as Virtual Private Clouds (VPC), subnets, route tables and firewalls. A basic VPC was set up on AWS with a public subnet for web servers and a private subnet for application and database servers. Route tables direct internet traffic through an internet gateway for the public subnet, while private resources reach the internet only through a NAT gateway. Security groups were configured to allow only the required ports, and secure communication

between instances was tested.

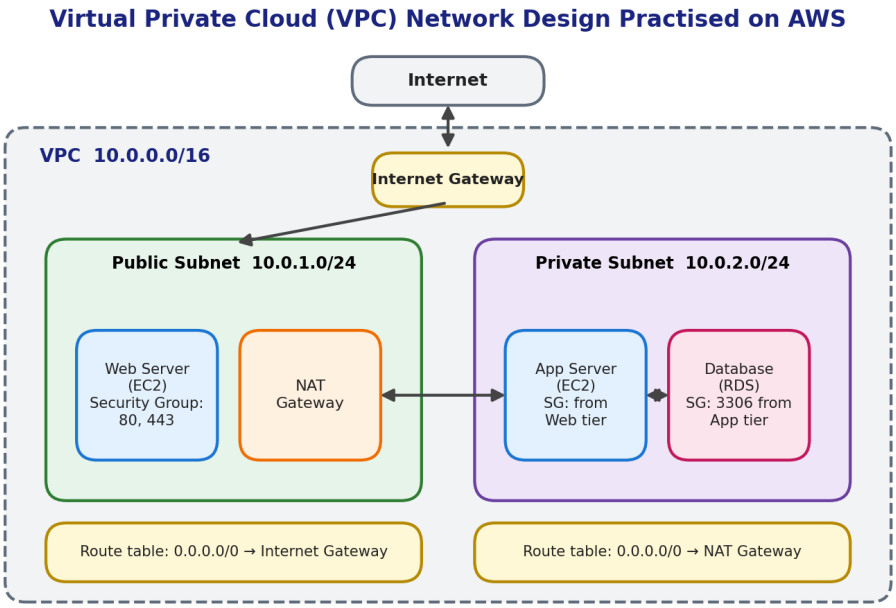


Figure 6.2: VPC with Public and Private Subnets

7. IDENTITY, SECURITY AND COMPLIANCE

7.1 Identity and Access Management (IAM)

On Day 6 identity and access management in cloud environments was studied. Users, groups and roles were created and permissions were granted through IAM policies written as JSON documents. The principle of least privilege—giving each identity only the permissions it needs—was applied, and multi-factor authentication (MFA) was enabled for added security.

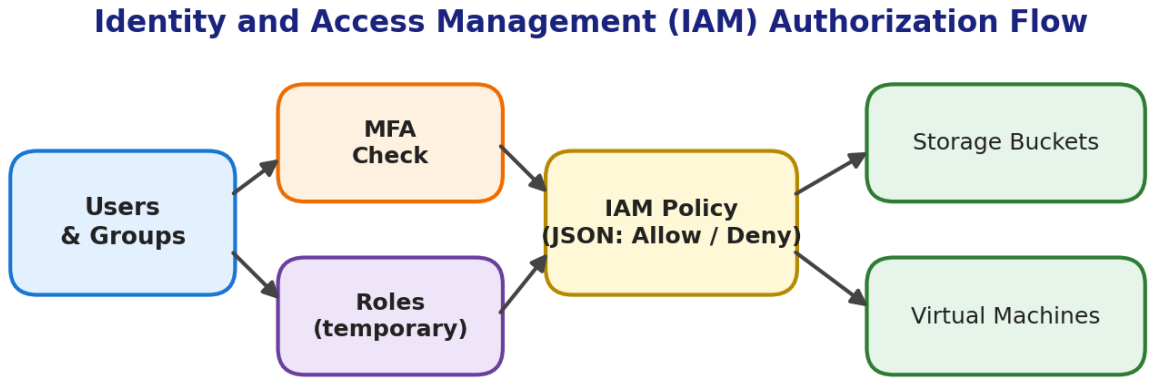


Figure 7.1: IAM Authorization Flow

7.2 Cloud Security Fundamentals

Cloud security fundamentals were covered on Day 13, beginning with the shared responsibility model. The provider secures the underlying infrastructure, while the customer remains responsible for securing data, identities, operating systems and network configuration. Basic security configurations were implemented using IAM and security groups, and tools and practices for data encryption and threat detection were studied.

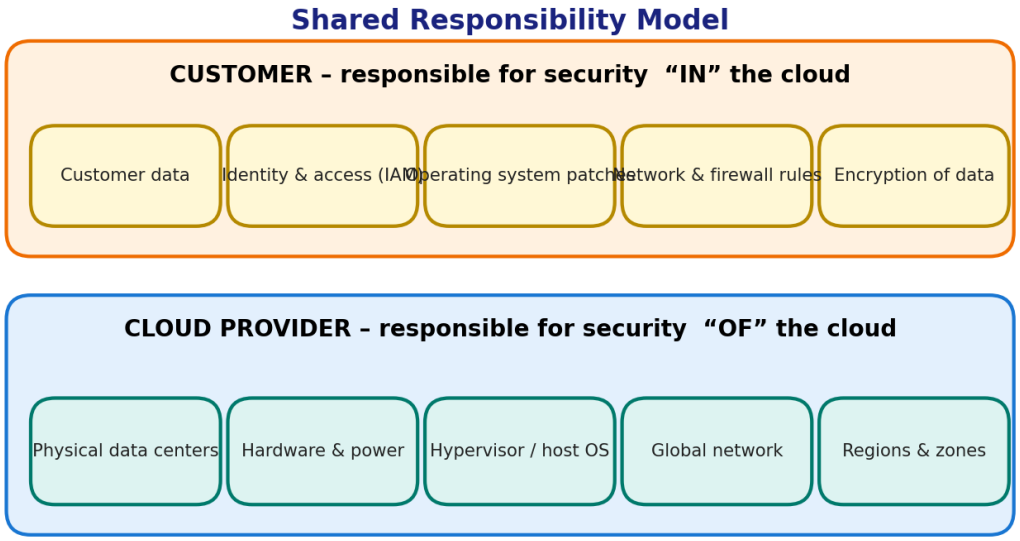


Figure 7.2: Shared Responsibility Model

7.3 Encryption and Key Management

Encryption and key management were explored on Day 16. Encryption was implemented for data at rest, using encrypted volumes and buckets, and for data in transit, using TLS. Best practices for protecting encryption keys were studied using AWS Key Management Service (KMS) and Azure Key Vault, including key rotation, separation of duties and restricting who can use each key.

7.4 Compliance, Governance and Risk Management

Compliance and risk management were the focus of Day 14 and were revisited on Day 23 as governance. Regulatory standards were studied together with the way cloud providers support them through auditing, logging and security controls. Risk assessment and the maintenance of audit trails were also practised.

Standard	Area	Cloud Support
GDPR	Protection of personal data of individuals in the EU	Region selection, encryption, access logs
HIPAA	Protection of health information	Access controls, audit trails, encryption
ISO 27001	Information security management systems	Provider certifications, security controls, monitoring

8. SCALABILITY, AVAILABILITY AND DISASTER RECOVERY

8.1 Auto-Scaling and Load Balancing

Day 7 focused on auto-scaling and load balancing. An auto-scaling group and an Elastic Load Balancer were configured on AWS so that compute capacity adjusts automatically to demand. When average CPU utilization rises above a threshold new instances are launched, and when demand falls the extra instances are removed. The load balancer spreads requests across healthy instances in different availability zones, which gives high availability and efficient use of resources.

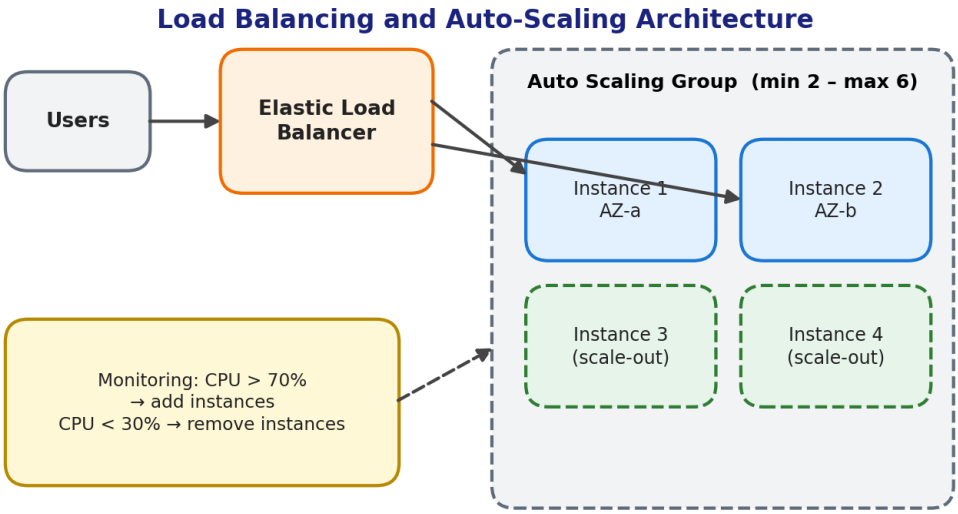


Figure 8.1: Load Balancer with Auto Scaling Group

8.2 Disaster Recovery and Business Continuity

Disaster recovery and business continuity planning in the cloud were studied on Day 22. Various backup strategies, automated failover mechanisms and multi-region deployment practices were examined with the aim of minimizing downtime and data loss. Two measures guide every plan: the Recovery Time Objective (RTO), which is how quickly service must be restored, and the Recovery Point Objective (RPO), which is how much data loss is acceptable.

Strategy	Description	Cost	Recovery Speed
Backup & restore	Periodic backups copied to another region	Lowest	Hours
Pilot light	Core components kept running at minimum size	Low	Tens of minutes
Warm standby	A scaled-down copy of the full system always running	Medium	Minutes
Multi-site active	Full capacity in more than one region	Highest	Near zero

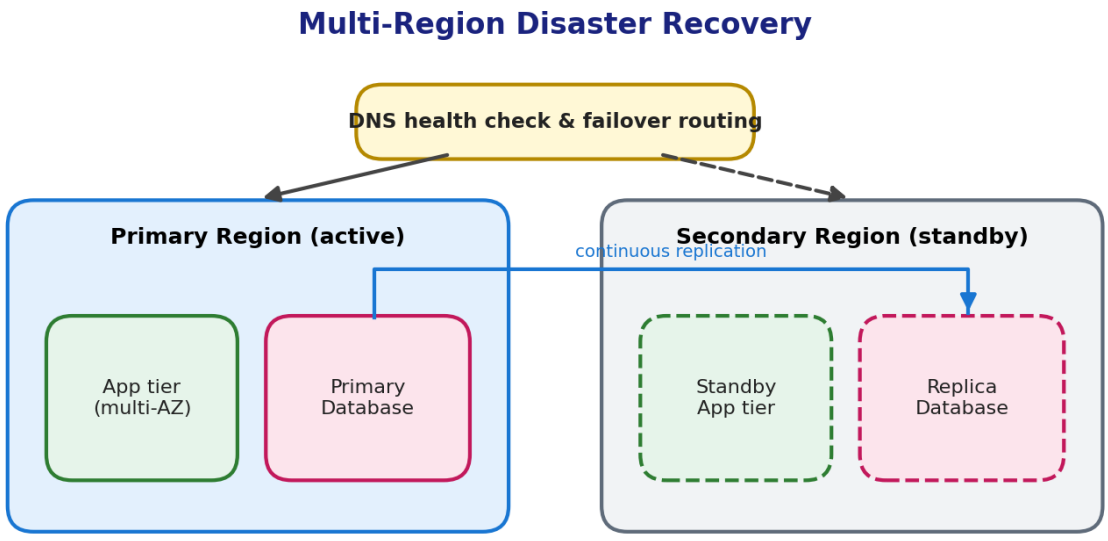


Figure 8.2: Multi-Region Disaster Recovery Architecture

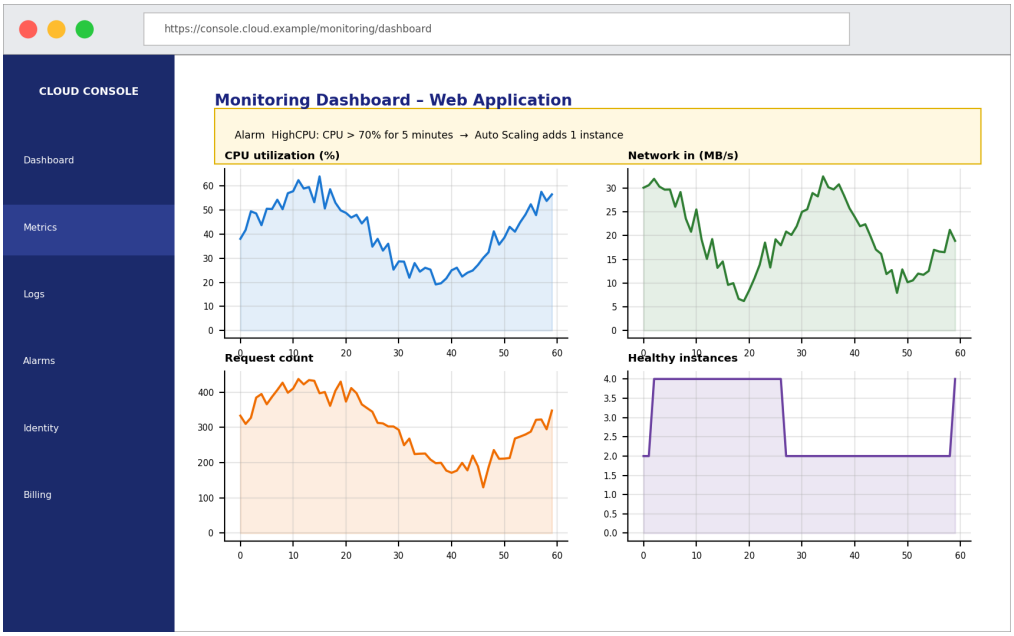


Figure 8.3: Monitoring Dashboard with Auto Scaling Alarm (representative console view)

9. PLATFORM SERVICES, DATABASES AND HYBRID CLOUD

9.1 Platform as a Service (PaaS)

Day 8 explored Platform as a Service offerings. A sample application was deployed using Heroku and Google App Engine. PaaS abstracts the management of servers, operating systems and runtimes, so a developer pushes code and the platform provisions, scales and monitors the application. The trade-off is less control over the underlying infrastructure than IaaS provides.

9.2 Database as a Service (DBaaS)

On Day 9 Database as a Service solutions such as Amazon RDS and Firebase were investigated. A relational database was set up in the cloud, its backup and recovery options were explored, and it was integrated with a simple application. Managed databases take over patching, backups, replication and failover, allowing developers to concentrate on the data model and queries.

Service	Type	Managed by Provider	Typical Use
Amazon RDS	Relational (MySQL, PostgreSQL, etc.)	Backups, patching, Multi-AZ failover	Transactional applications

Service	Type	Managed by Provider	Typical Use
Firebase	NoSQL real-time database	Scaling, sync, hosting	Mobile and web apps
BigQuery / Redshift	Analytical warehouse	Storage, compute, scaling	Reporting and analytics

9.3 Service-Model Review and Provider Case Study

On Day 10 the service models were reviewed and use cases for hybrid and multi-cloud strategies were explored. A small case study was started that compared the features and pricing of different cloud providers for a sample application scenario, which showed that the best choice depends on workload, skills, compliance needs and cost.

9.4 Hybrid Cloud Solutions

Hybrid cloud solutions were studied on Day 12 to understand how on-premises infrastructure integrates with cloud services. The benefits—using existing investments, keeping sensitive data local and gaining elastic capacity—were weighed against the challenges of connectivity, consistent security and managing data and applications across different environments. Hybrid architectures were designed using a secure VPN or dedicated link between the data center and the cloud network.

10. CLOUD COST MANAGEMENT

10.1 Pricing Models

Cloud cost management was studied on Day 11 to learn how to optimize and control cloud expenditure. The pricing models offered by providers were explored, and cost management tools were used to track and analyze spending.

Pricing Model	How It Works	Best Suited For
Pay-as-you-go (on-demand)	Pay per second or hour with no commitment	Unpredictable or short-term workloads
Reserved instances	Commit to one or three years for a lower rate	Steady, long-running workloads

Pricing Model	How It Works	Best Suited For
Spot / preemptible	Use spare capacity at a large discount; may be interrupted	Batch jobs, fault-tolerant tasks
Free tier	Limited free usage for learning	Experiments and training labs

10.2 Cost Monitoring and Optimization

The later part of the internship (Day 26) returned to cost management and optimization using AWS Cost Explorer, Azure Cost Management and Google Cloud Cost Management. Cost reports were created, budget alerts were set, and best practices for reducing cloud costs were applied.

- **Right-sizing:** choose instance sizes that match actual CPU and memory use.
- **Storage optimization:** move rarely used data to cheaper tiers and delete unused volumes and snapshots.
- **Cost-effective services:** select the most economical service for each task, such as serverless for occasional workloads.
- **Budgets and alerts:** receive notifications before spending exceeds the planned limit.



Figure 10.1: Cost Explorer with Budget Alert (representative console view)

11. DEVOPS, CI/CD AND INFRASTRUCTURE AS CODE

11.1 Cloud-Based DevOps and CI/CD

Day 15 focused on cloud-based DevOps practices. Continuous Integration and Continuous Deployment (CI/CD) pipelines were learned and tools like GitHub Actions and AWS CodePipeline were used to automate application deployment. This gave hands-on experience with version control, build automation and code deployment: every commit triggers an automated build and test, and a successful run is released to the cloud without manual steps.

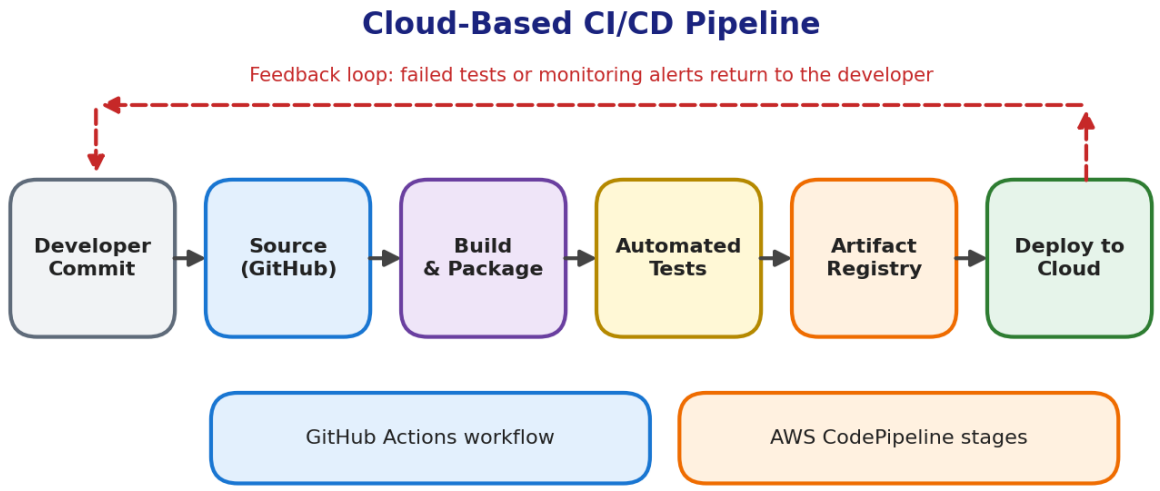


Figure 11.1: Cloud-Based CI/CD Pipeline

11.2 Infrastructure as Code (IaC)

On Day 20 Infrastructure as Code concepts and tools such as Terraform and AWS CloudFormation were studied. Infrastructure is defined in text files, reviewed and stored in version control, and then provisioned automatically. This makes environments reproducible, supports collaboration and removes the errors of manual configuration.

Benefit	Explanation
Reproducibility	The same code creates identical development, test and production environments
Version control	Changes are tracked, reviewed and can be rolled back
Speed	Complete environments are created in minutes

Benefit	Explanation
Collaboration	Teams share and improve one definition of the infrastructure

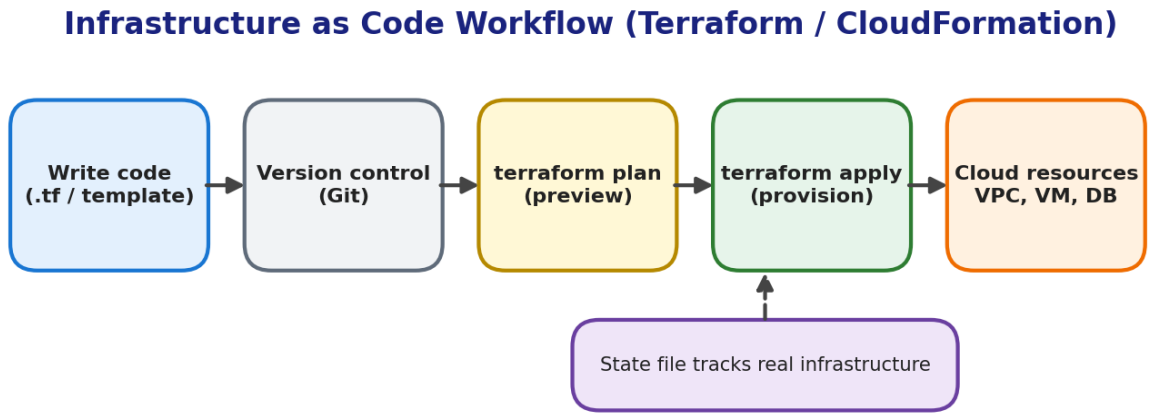


Figure 11.2: Infrastructure as Code Workflow

12. CLOUD-NATIVE: MICROSERVICES, KUBERNETES, SERVERLESS

12.1 Cloud-Native Applications and Microservices

Day 17 was spent learning about cloud-native applications and microservices architecture. Applications built specifically for cloud environments are divided into small independent services that can be developed, deployed and scaled separately. The focus was on scalability, resilience and efficient resource utilization, and container orchestration with Kubernetes was explored.

12.2 Container Orchestration with Kubernetes

Cloud orchestration and automation were studied on Day 19, particularly Kubernetes. Containerized applications were deployed, managed and scaled within a Kubernetes cluster, which improved knowledge of high-availability deployment. The control plane stores the desired state and schedules pods onto worker nodes, restarting any pod that fails and adding replicas on request.

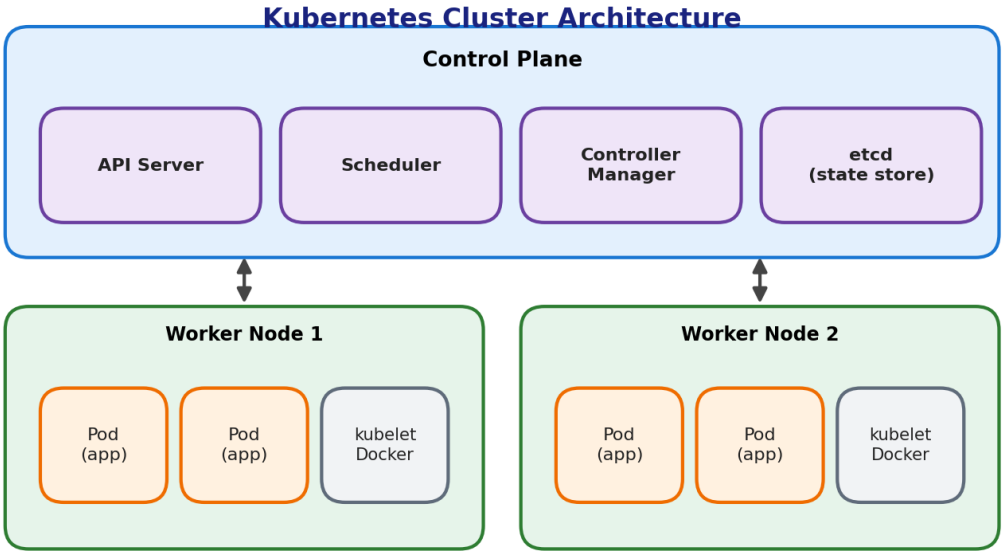


Figure 12.1: Kubernetes Cluster Architecture

```
Terminal – kubectl

$ kubectl get nodes
NAME          STATUS   ROLES           AGE   VERSION
node-1        Ready    control-plane   3d    v1.29
node-2        Ready    <none>          3d    v1.29

$ kubectl scale deployment web-app --replicas=4
deployment.apps/web-app scaled
$ kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
web-app-7d9f8c6b5-4kx2p   1/1     Running   0          2m
web-app-7d9f8c6b5-9qm7z   1/1     Running   0          2m
web-app-7d9f8c6b5-lt5vn   1/1     Running   0          14s
web-app-7d9f8c6b5-xw8rj   1/1     Running   0          14s
```

Figure 12.2: Scaling a Deployment with kubectl (representative terminal session)

12.3 Serverless Computing and Functions-as-a-Service

Serverless computing and Functions-as-a-Service (FaaS) were explored on Day 24 using AWS Lambda and Azure Functions. Event-driven applications were built and deployed, in which a function runs only when an event such as an HTTP request or a file upload occurs. This gives cost-effective and automatically scalable application development because there are no idle servers to pay for.

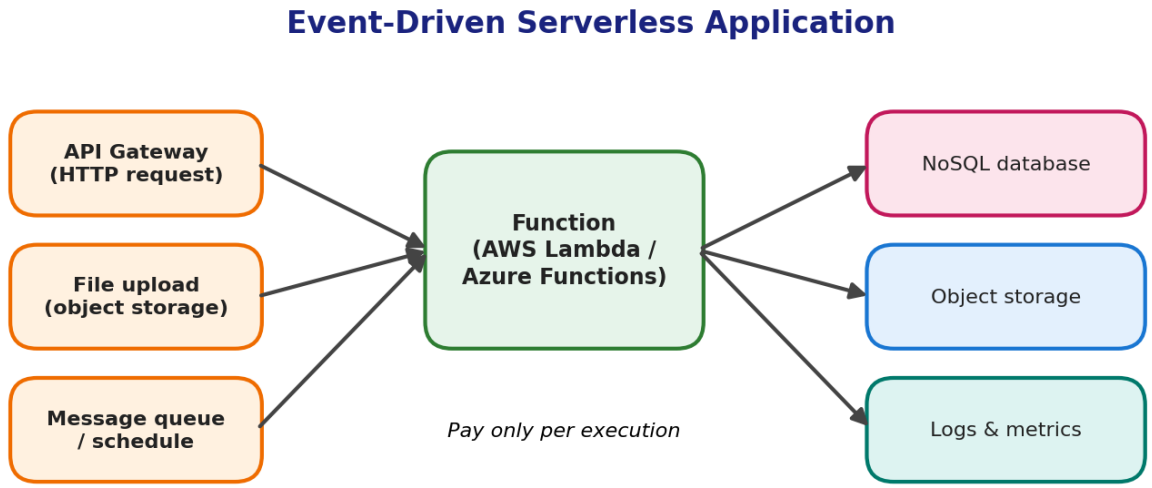


Figure 12.3: Event-Driven Serverless Application

13. MIGRATION, MONITORING AND LOGGING

13.1 Cloud Migration Strategies

Day 18 explored cloud migration strategies: how to plan and execute the migration of applications and data from on-premises environments to the cloud. Tools and methodologies such as AWS Migration Hub and Azure Migrate were studied and migration plans were developed.

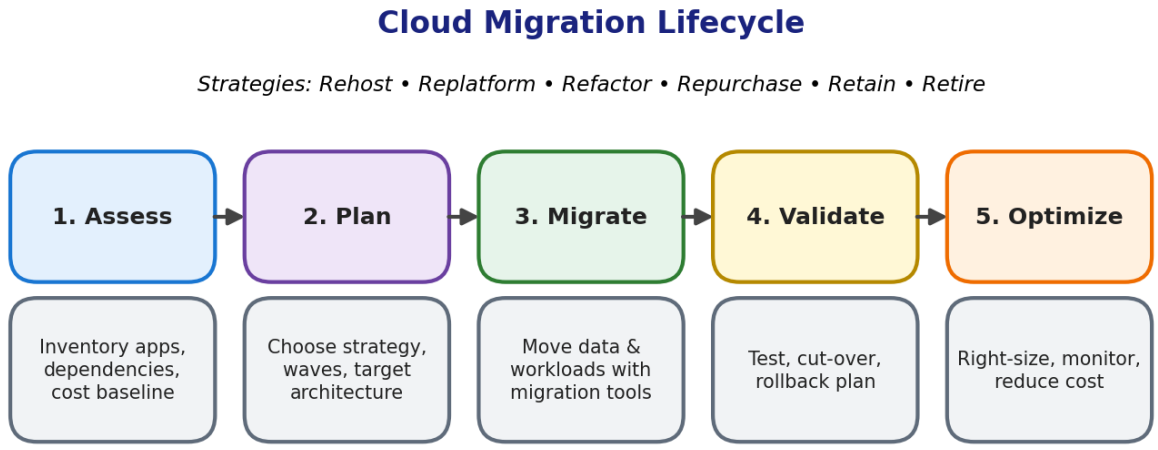


Figure 13.1: Cloud Migration Lifecycle

Strategy	Meaning	Effort
Rehost	Move servers as they are (“lift and shift”)	Low

Strategy	Meaning	Effort
Replatform	Make small optimizations, such as moving to a managed database	Medium
Refactor	Redesign the application to be cloud-native	High
Repurchase / Retire / Retain	Switch to SaaS, switch off, or keep on-premises	Varies

13.2 Monitoring and Logging

On Day 21 monitoring and logging in cloud environments was the focus. Tools like Prometheus, Grafana and AWS CloudWatch were explored to track system performance, detect anomalies and ensure high availability of cloud-based services. Metrics describe how a system is performing, logs record what happened, and alarms notify the team or trigger automatic actions such as scaling.

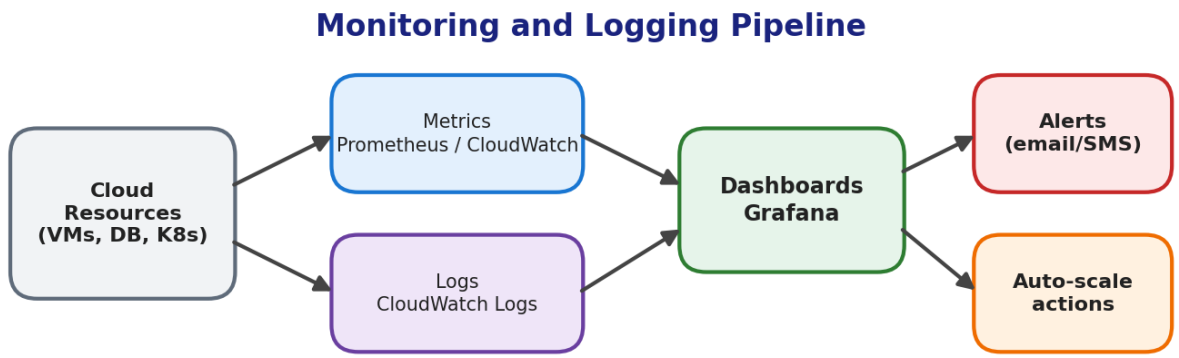


Figure 13.2: Monitoring and Logging Pipeline

14. APIS, DATA WAREHOUSING AND APPLICATION FRAMEWORKS

14.1 Cloud API Management

On Day 27 the focus was cloud API management: how to create, deploy and manage APIs in a cloud environment. Best practices for securing and scaling APIs were studied. An API gateway acts as a single entry point that authenticates callers, limits request rates, caches responses and records logs before requests reach the backend

services.

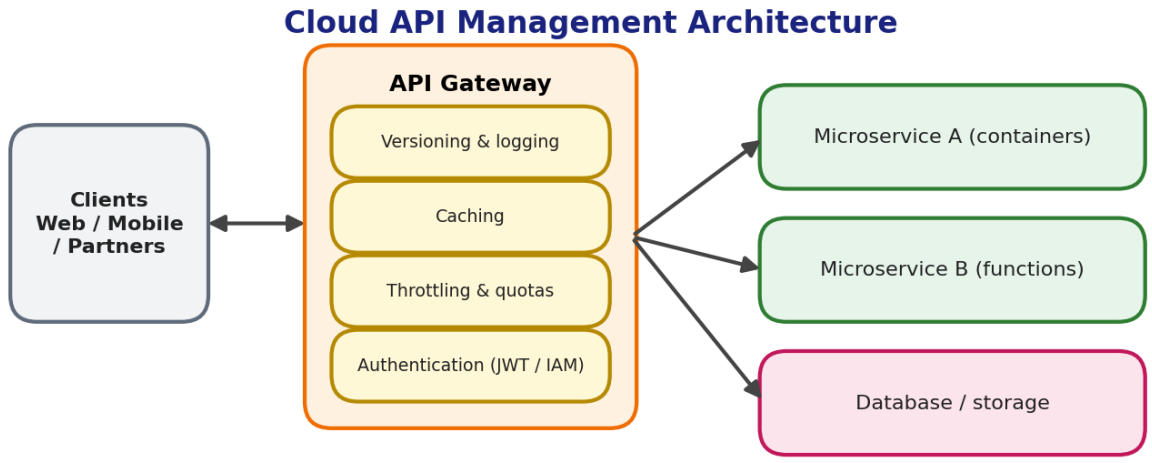


Figure 14.1: Cloud API Management Architecture

14.2 Cloud Data Warehousing and ETL

Day 28 explored cloud-based data warehousing and ETL (Extract, Transform, Load) processes. Data pipelines were designed and implemented using cloud services, focusing on data integration, transformation and loading techniques. Tools like AWS Redshift and Google BigQuery were used for data warehousing tasks.

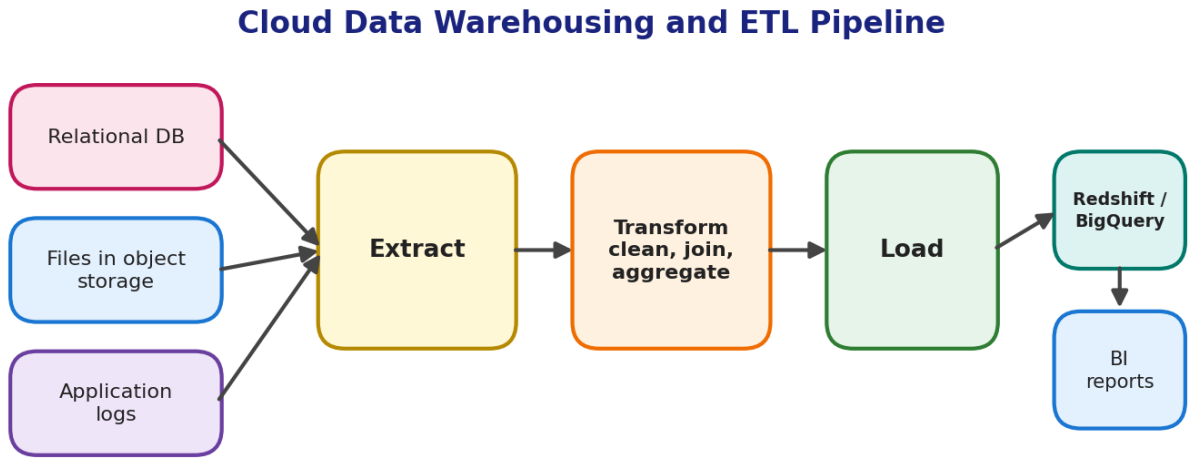


Figure 14.2: Data Warehousing and ETL Pipeline

14.3 Cloud Application Development Frameworks

Day 29 was spent on cloud-based application development frameworks. Frameworks provided by cloud platforms were studied to accelerate application development and deployment, including AWS Amplify and Azure App Service. Applications were built

and deployed using these tools.

Framework	Provider	Purpose
AWS Amplify	Amazon Web Services	Build and host full-stack web and mobile apps with authentication, storage and APIs
Azure App Service	Microsoft Azure	Host web apps and APIs with built-in scaling and deployment slots

15. MINI PROJECT AND CAPSTONE PROJECT

15.1 Mini Project: Scalable Web Application Architecture

On Day 25 a mini project was carried out to design a cloud-based architecture for a scalable web application. Load balancers, auto-scaling groups, storage solutions and security components were incorporated into the design, and the architecture was documented.

15.2 Capstone Project

On the final day (Day 30) knowledge was consolidated by reviewing all the concepts and tools learned during the internship. A capstone project was completed in which a comprehensive cloud solution was designed and implemented, applying everything learned over the 30 days.

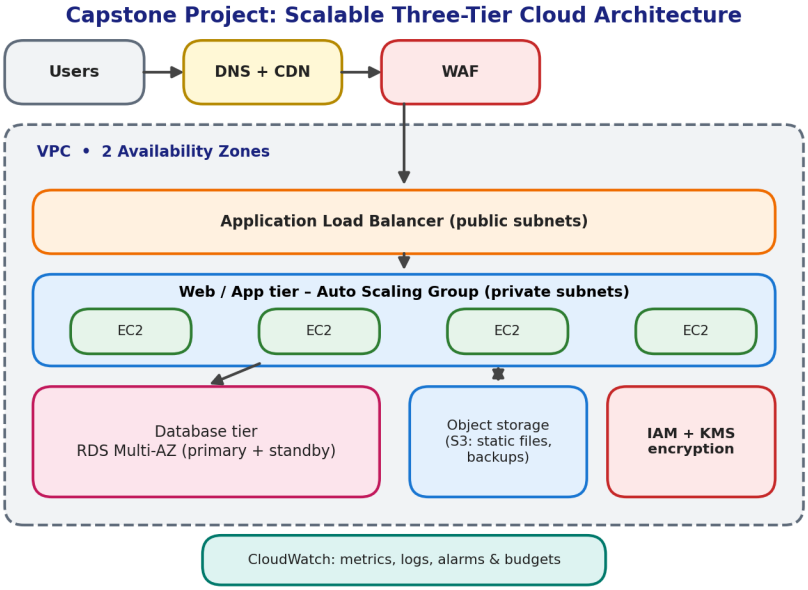


Figure 15.1: Scalable Three-Tier Cloud Architecture (Capstone Design)

15.3 Design Decisions in the Capstone

Requirement	Design Decision	Service Used
High availability	Two availability zones and a load balancer	Elastic Load Balancer, Auto Scaling
Scalability	Add or remove web instances based on CPU load	Auto Scaling Group
Security	Private subnets, least-privilege roles, encrypted data	VPC, IAM, KMS, security groups
Durability	Multi-AZ database and versioned object storage	RDS, S3
Observability	Central metrics, logs and alarms	CloudWatch
Cost control	Right-sized instances, budgets and alerts	Cost Explorer, Budgets

16. LEARNINGS AND OUTCOMES

16.1 Learnings from the Internship

- **Gain experience:** Job listings often state that they prefer candidates with educational and job experience. If you're new to the workforce or attending school, you may consider looking for an internship to gain the experience required for most entry-level positions.
- **Identify career goals:** An internship can give you an authentic experience in a job role by providing you with an introductory experience to a career path, its duties and daily operations. If you enjoy your internship, this might indicate that your career is on the right path.
- **Strengthen a resume:** Internships can give you workplace experience before you actually enter the workforce. They also may assist you with developing additional skills to list on your resume, which can emphasize your value as a candidate.
- **Attain college credits:** Some internship opportunities may offer you college credit for your time as an intern. An internship that offers you both college credit and experience can be ideal for those who are looking to graduate with work

experience.

16.2 Technical Skills Gained

Area	Skills Developed	Evidence from the Internship
Cloud fundamentals	Service and deployment models, virtualization	Days 1 – 2, 10
Infrastructure	VMs, containers, storage, VPC networking	Days 3 – 5
Security	IAM, MFA, encryption, compliance	Days 6, 13, 14, 16, 23
Reliability	Auto-scaling, load balancing, disaster recovery	Days 7, 22
Automation	CI/CD, Infrastructure as Code, Kubernetes	Days 15, 19, 20
Cloud-native	Microservices, serverless, API management	Days 17, 24, 27
Data & cost	DBaaS, data warehousing, cost optimization	Days 9, 11, 26, 28

16.3 Professional Skills

Beyond technical skills, the internship strengthened the habits of planning work in small steps, documenting architectures clearly, comparing alternatives before choosing a service and learning independently from provider documentation.

17. CONCLUSION AND REFERENCES

17.1 Conclusion

Throughout the 30-day Cloud Computing internship, I gained a comprehensive understanding of cloud technologies and their applications. This journey provided me with hands-on experience in various facets of cloud computing, from fundamental concepts such as virtualization and cloud architectures to advanced topics like serverless computing, cloud security, and multi-cloud strategies. I explored a wide

range of tools and services across different cloud platforms, including AWS, Azure, and Google Cloud, and developed practical skills in managing cloud resources, optimizing costs, and implementing robust security measures.

My experience included working on real-world scenarios such as disaster recovery, cloud migration, and developing cloud-native applications. This internship has significantly enhanced my technical expertise and prepared me for a future career in cloud computing, equipping me with the skills and knowledge to tackle complex challenges in the cloud domain.

17.2 References

1. Amazon Web Services, AWS Documentation, https://docs.aws.amazon.com/
2. Microsoft, Azure Documentation, https://learn.microsoft.com/azure/
3. Google Cloud, Google Cloud Documentation, https://cloud.google.com/docs
4. Kubernetes Authors, Kubernetes Documentation, https://kubernetes.io/docs/
5. HashiCorp, Terraform Documentation, https://developer.hashicorp.com/terraform/docs
6. Docker Inc., Docker Documentation, https://docs.docker.com/